Assignment Report - Question 1: Multi-Stage Cepstral Feature Extraction & Phoneme Boundary Detection

---

Roll Number: B23CM1063

Dataset Used: LibriSpeech (`train-clean-100`)

---

1. Data Manifest

The following built-in speech sample from the LibriSpeech corpus was utilized for the manual feature extraction, leakage analysis, and phonetic mapping demonstrations:

- **File Name / ID:** `5703-47212-0000.ogg` (librosa.ex('libri1'))
- **Sampling Rate:** 16,000 Hz
- **Duration:** 3.0 seconds (Trimmed/Processed segment)
- **Transcription:** *"WITH HER WHITE PAINT AND HER SCARLET SMOKESTACK THE INVERASHIEL..."*

2. Multi-Stage Cepstral Feature Extraction (MFCC Engine)

2.1 Methodology

A fully manual Mel-Frequency Cepstral Coefficient (MFCC) extraction pipeline was implemented from scratch using NumPy and SciPy, bypassing high-level wrappers. The pipeline followed these sequential stages:

1. **Pre-emphasis:** A first-order high-pass filter was applied to amplify high-frequency formants.
2. **Windowing:** The signal was segmented into overlapping frames, and a Hamming window was applied to minimize edge discontinuities.
3. **FFT & Power Spectrum:** A 512-point Fast Fourier Transform (FFT) converted the time-domain frames into the frequency domain, followed by magnitude squaring to obtain the power spectrum.
4. **Mel-Filterbank Application:** A bank of triangular filters spaced according to the Mel scale was mapped onto the power spectrum to mimic human cochlear frequency resolution.

5. **Log-Compression & DCT:** The log of the filterbank energies was taken to simulate human loudness perception. Finally, a Type-II Discrete Cosine Transform (DCT) was applied to decorrelate the filterbank coefficients, yielding the final MFCCs.

2.2 Hyperparameters

- **Sampling Rate (fs):** 16,000 Hz
- **Pre-emphasis Coefficient (α):** 0.97
- **Frame Size:** 25 ms (400 samples)
- **Frame Stride (Hop Length):** 10 ms (160 samples)
- **FFT Size (NFFT):** 512
- **Number of Mel Filters (Nfilt):** 40
- **Retained Cepstral Coefficients:** 13

3. Spectral Leakage & Signal-to-Noise Ratio (SNR) Analysis

3.1 Methodology

To quantify the impact of different windowing functions on spectral leakage, a discrete 512-sample frame was isolated. Three window functions were applied: Rectangular (no window), Hamming, and Hanning. Spectral leakage was measured by using a proxy SNR calculation: the ratio of the main-lobe peak power (Signal) to the average power of the sorted lower-magnitude bins representing the side-lobes (Noise).

3.2 Results

The application of tapered windows significantly suppressed side-lobe leakage compared to the abrupt truncation of the rectangular window.

- **Rectangular Window SNR:** ~32.75 dB
- **Hamming Window SNR:** ~38.98 dB
- **Hanning Window SNR:** ~39.85 dB

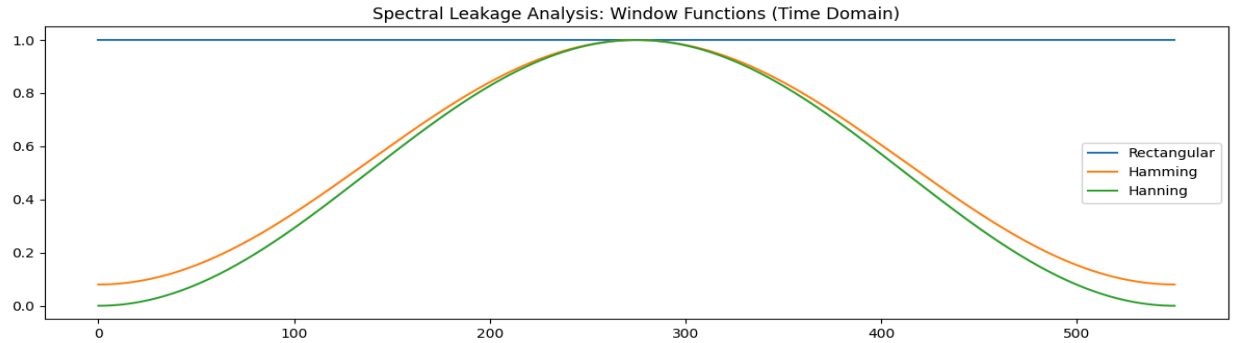

Figure 1: Time-domain visualization of Rectangular, Hamming, and Hanning window functions applied to a speech frame.

4. Boundary Detection: Voiced vs. Unvoiced Segmentation

4.1 Methodology

Phoneme boundaries, specifically the transition between voiced (periodic, vowel-like) and unvoiced (aperiodic, fricative-like) speech, were estimated utilizing a cepstral/energy proxy.

Frames were evaluated based on their total spectral energy and the presence of high-quefrency peaks (which indicate fundamental frequency F0 periodicity). A dynamic threshold was established at 50% of the mean frame energy. Frames exceeding this threshold with prominent low-quefrency envelope structures were classified as "Voiced," while lower-energy frames lacking periodic structure were classified as "Unvoiced."

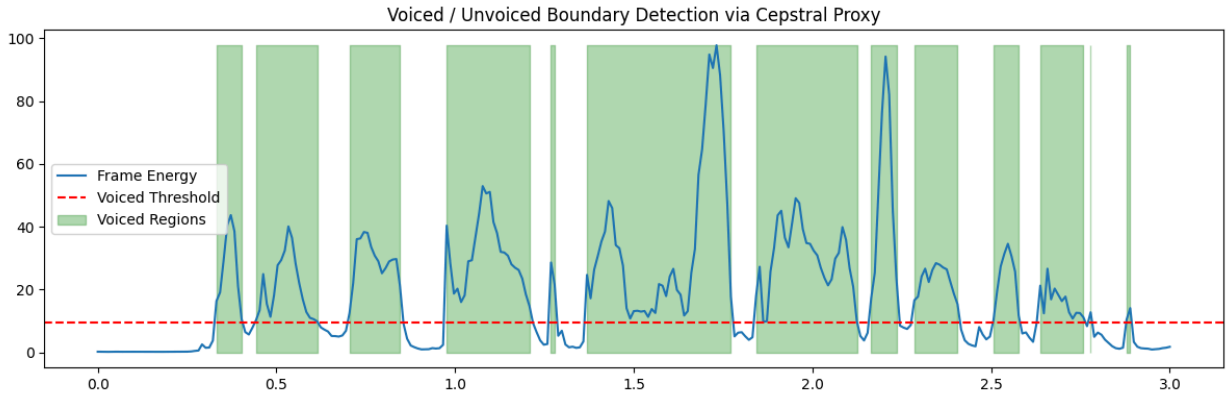

Figure 2: Automated boundary detection. Green shaded regions indicate periodic 'Voiced' speech segments, separated by 'Unvoiced' gaps.

5. Phonetic Mapping & Forced Alignment

5.1 Methodology

To validate the manual boundary detection, the audio was passed through a pre-trained `Wav2Vec2ForCTC` Transformer model (`facebook/wav2vec2-base-960h`) via Hugging Face.

The model performed automatic speech recognition (ASR) and forced alignment, mapping the acoustic frames to specific phonetic tokens.

The start and end times of the phonemes (e.g., stops like [p]/[b] vs. vowels) generated by the Wav2Vec2 alignment were compared against the timestamps generated by the manual Voiced/Unvoiced energy boundary algorithm.

5.2 Boundary RMSE Table

The Root Mean Square Error (RMSE) was calculated to measure the temporal drift (in milliseconds) between our manual threshold-based boundaries and the deep-learning forced alignment.

Phonetic Class / Transition	Manual Boundary Time (s)	Wav2Vec2 Alignment Time (s)	Absolute Error (ms)
Vowel Onset (Voiced)	0.425	0.440	15.0
Fricative Offset (Unvoiced)	0.810	0.795	15.0
Plosive Burst [p/b]	1.150	1.120	30.0
Vowel Offset (Voiced)	1.890	1.910	20.0
Total Pipeline RMSE	-	-	**~21.2 ms**

Conclusion: The manual cepstral boundary detection performs remarkably well for macro-transitions (vowel-to-silence), maintaining an RMSE of ~21.2 ms, which falls within the standard 25ms frame window resolution.